

A Poisson process  $N_t, t \geq 0$  is defined by its instantaneous jump probability *now* and is dragged forward by independence:

**Definition 0.1:**  $N$  is a Poisson process with parameter  $\lambda > 0$  if both of these conditions are satisfied:

1. (Generator) the time evolution of the probabilities  $p_k(t) := \mathbb{P}(N_t = k)$  satisfies

$$p'_k(0) = \begin{cases} \lambda & \text{if } k = 1 \\ -\lambda & \text{if } k = 0 \\ 0 & \text{if } k > 1 \end{cases}$$

2. (Drag forward)  $N$  has independent, i.e. each increment  $\Delta N_{t,h} := N_{t+h} - N_t$  is independent from  $N_t$  (denoted  $\Delta N_{t,h} \perp N_t$ ), i.e. satisfies

$$\mathbb{P}(\Delta N_{t,h} = k \text{ and } N_t = n) = \mathbb{P}(\Delta N_{t,h} = k) \mathbb{P}(N_t = n)$$

for all  $n, k$ .

3.  $N$  has stationary increments, i.e. all increments  $\Delta N_{\cdot,h}$  have the same laws, i.e.

$$\Delta N_{t,h} \sim \Delta N_{0,h} \text{ for all } t \geq 0.$$

To drag the probabilities forward, we choose a later time  $t \geq 0$  and apply the conditions above to

$$\begin{aligned} \mathbb{P}(N_{t+h} = k) &= \sum_n \mathbb{P}(N_{t+h} = k \mid N_t = n) && \text{(by total probability)} \\ &= \sum_n \mathbb{P}(\Delta N_{t,h} = k - n \mid N_t = n) && \text{(by } N_{t+h} = N_t + \Delta N_{t,h}) \\ &= \sum_n \mathbb{P}(\Delta N_{t,h} = k - n) \cdot \mathbb{P}(N_t = n) && \text{(by } \Delta N_{t,h} \perp N_t) \\ &= \sum_n \mathbb{P}(\Delta N_{0,h} = k - n) \cdot \mathbb{P}(N_t = n) && \text{(by } \Delta N_{t,h} \sim \Delta N_{0,h}) \\ &= \sum_n \mathbb{P}(N_h = k - n) \cdot \mathbb{P}(N_t = n) && \text{(by } N_t = N_0 + \Delta N_{0,h}) \end{aligned}$$

In terms of the functions  $p_k(t)$ , this means

$$p_k(t+h) = \sum_n p_n(h) p_{k-n}(t)$$

for all  $k, t, h$ . Differentiate w.r.t.  $h$ :

$$p'_k(t+h) = \sum_n p'_n(h) p_{k-n}(t)$$

and evaluate at  $h = 0$ :

$$p'_k(t) = \sum_n p'_n(0) p_{k-n}(t)$$

This means:

| for $k = 0$                                                                                                                                   | for $k > 0$                                                                                                                                             |
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| in this case, the sum has a single term only,<br>$p'_0(t) = p'_0(0)p_0(t),$ and with $p'_0(0) = -\lambda$ this means<br>$p'_0 = \lambda p_0.$ | $p'_n(0)$ was postulated to vanish for $n > 1$ , so the sum is just<br>$p'_k(t) = p'_0(0)p_k(t) + p'_1(0)p_{k-1}(t)$ $= \lambda(-p_k(t) + p_{k-1}(t)),$ |

or, in point-free style,

$$p'_0 = \lambda p_0$$

$$p'_k = \lambda(p_{k-1} - p_k) \text{ for } k > 0.$$

## Computing $p_k$

Collect all  $p_k$  in the generating function

$$G_t(x) = \sum_k p_k(t)x^k$$

and differentiate

$$\begin{aligned} \partial_t G_t(x) &= \sum_k p'_k(t)x^k \\ &= p'_0(t) + \sum_{k \geq 1} p'_k(t)x^k \\ &= p'_0(t) + \lambda \sum_{k \geq 1} p_{k-1}(t)x^k - \lambda \sum_{k \geq 1} p_k(t)x^k \\ &= p'_0(t) + \lambda x G_t(x) - \lambda(G_t(x) - p_0(t)) \\ &= \lambda(x-1)G_t(x), \end{aligned}$$

so

$$\frac{d}{dt}G_t(x) = \lambda(x-1)G_t(x).$$

The solution is

$$G_t(x) = \exp(\lambda(x-1)t).$$

To get the coefficients of  $G_t(x)$ , expand

$$\begin{aligned} \exp(\lambda(x-1)t) &= e^{-\lambda t} \exp(\lambda t x) = e^{-\lambda t} \sum_k \frac{(\lambda x t)^k}{k!} \\ &= \sum_k e^{-\lambda t} \frac{(\lambda t)^k}{k!} x^k \end{aligned}$$

Equating with  $G_t(x) = \sum_k p_k(t)x^k$  we get

$$p_k(t) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}$$